VNR VJIET

Name of the Laboratory:

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

WEEK-6

Implementation of the following Page Replacement Algorithms a) FIFO b) LRU c) optimal

AIM- To Implement of the Page Replacement Algorithms such as FIFO, LRU, optimal

ALGORITHM-

FIFO:

1. Start.
2. Read the number of pages in the reference string.
3. Read the reference string.
4. Read the number of frames.
5. Initially mark all frames as empty.
6. For each page in the reference string:
7. Check whether the page is already present in a frame.
8. If present, it is a page hit.
9. If not present, it is a page fault.
10. If there is an empty frame, insert the page.
11. If all frames are full, replace the page that entered first.
12. Move the FIFO pointer to the next frame.
13. Display the page faults.
14. Stop.

LRU:

- Start.
- Read the reference string.
- Read the number of frames.
- Initially mark all frames as empty.
- For each page:
- Check whether the page is already present.
- If present, it is a **hit**.
- If not present, it is a **page fault**.
- If an empty frame is available, insert the page.
- If all frames are full, find the page that has **not been used for the longest time**.
- Replace that page with the new page.

- Continue until all pages are processed.
- Display total page faults.
- Stop.

OPTIMAL:

- Start.
- Read the reference string.
- Read the number of frames.
- Initially mark all frames as empty.
- For each page:
- Check whether the page is already present.
- If present, it is a **hit**.
- If not present, it is a **page fault**.
- If an empty frame is available, insert the page.
- If all frames are full:
- Look at the **future references**.
- Find which page in the frames will be used **farthest in the future**.
- Replace that page.
- Continue until all pages are processed.
- Display total page faults.
- Stop.

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CODE:

```
#include <stdio.h>
int main()
{
int pages[50],frames[10];
int n,f;
int i,j,k;
int pointer=0;
int pageFaults=0;
int found;
printf("Enter number of pages: ");
scanf("%d",&n);
printf("Enter reference string:\n");
for(i=0;i<n;i++)
scanf("%d",&pages[i]);
printf("Enter number of frames: ");
scanf("%d",&f);
for(i=0;i<f;i++)
frames[i]=-1;
printf("\nFIFO Page Replacement\n");
for(i=0;i<n;i++)
{
found=0;
for(j=0;j<f;j++)
{
if(frames[j]==pages[i])
```

```
{
found=1;
break;
}
}
if(found==0)
{
frames[pointer]=pages[i];
pointer=(pointer+1)%f;
pageFaults++;
}
printf("%d : ",pages[i]);
for(k=0;k<f;k++)
{
if(frames[k]==-1)
printf("- ");
else
printf("%d ",frames[k]);
}
if(found==0)
printf("Page Fault");
else
printf("Hit");
printf("\n");
}
printf("\nTotal Page Faults = %d\n",pageFaults);
return 0;
}
```

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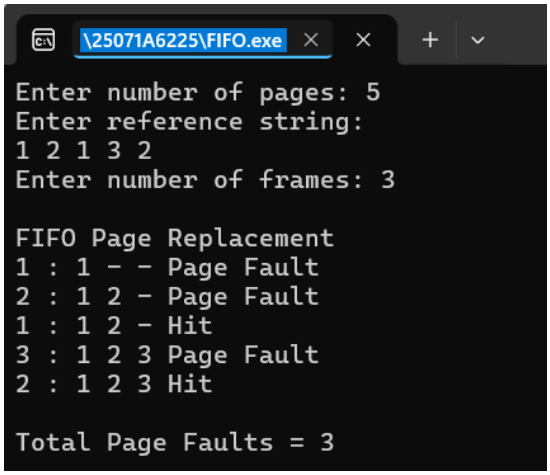
Name of the Laboratory:

Name of the Experiment:_____

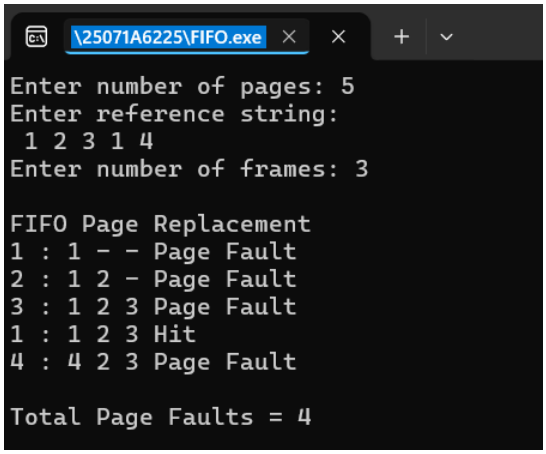
Experiment No:_____ **Date:**_____

OUTPUT:

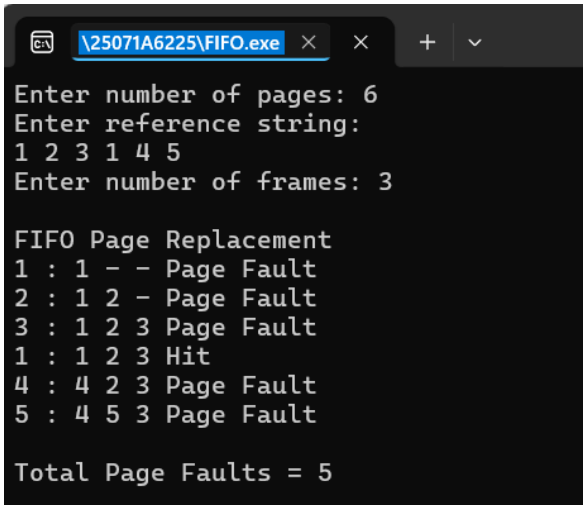
TEST CASE 1:



TEST CASE 2:



TEST CASE 3:



CODE:

```
#include <stdio.h>
int main()
{
int pages[50],frames[10],recent[10];
int n,f;
int i,j,k;
int found,pos;
int pageFaults=0;
int time=0;
int least;
printf("Enter number of pages: ");
scanf("%d",&n);
printf("Enter reference string:\n");
for(i=0;i<n;i++)
scanf("%d",&pages[i]);
printf("Enter number of frames: ");
scanf("%d",&f);
for(i=0;i<f;i++)
{
frames[i]=-1;
recent[i]=0;
}
printf("\nLRU Page Replacement\n");
for(i=0;i<n;i++)
{
found=0;
time++;
for(j=0;j<f;j++)
```

```
{
if(frames[j]==pages[i])
{
found=1;
recent[j]=time;
break;
}
}
if(found==0)
{
pageFaults++;
pos=-1;
for(j=0;j<f;j++)
{
if(frames[j]==-1)
{
pos=j;
break;
}
}
if(pos==-1)
{
least=recent[0];
pos=0;
for(j=1;j<f;j++)
{
```

```
if(recent[j]<least)
{
least=recent[j];
pos=j;
}
}
}
frames[pos]=pages[i];
recent[pos]=time;
}
printf("%d : ",pages[i]);
for(k=0;k<f;k++)
{
if(frames[k]==-1)
printf("- ");
else
printf("%d ",frames[k]);
}
if(found==0)
printf("Page Fault");
else
printf("Hit");
printf("\n");
}
printf("\nTotal Page Faults = %d\n",pageFaults);
return 0;
}
```

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OUTPUT:

TEST CASE 1:

```
s\25071A6225\LRU.exe
Enter number of pages: 5
Enter reference string:
 1 2 3 2 4
Enter number of frames: 3

LRU Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
3 : 1 2 3 Page Fault
2 : 1 2 3 Hit
4 : 4 2 3 Page Fault

Total Page Faults = 4
```

TEST CASE 2:

```
s\25071A6225\LRU.exe
Enter number of pages: 5
Enter reference string:
 1 2 1 3 2
Enter number of frames: 3

LRU Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
1 : 1 2 - Hit
3 : 1 2 3 Page Fault
2 : 1 2 3 Hit

Total Page Faults = 3
```

TEST CASE 3:

```
s\25071A6225\LRU.exe
Enter number of pages: 6
Enter reference string:
1 2 3 1 4 2
Enter number of frames: 3

LRU Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
3 : 1 2 3 Page Fault
1 : 1 2 3 Hit
4 : 1 4 3 Page Fault
2 : 1 4 2 Page Fault

Total Page Faults = 5
```

CODE:

```
#include <stdio.h>
int main()
{
int pages[50],frames[10];
int n,f;
int i,j,k;
int found,pos;
int pageFaults=0;
int farthest,nextUse;
printf("Enter number of pages: ");
scanf("%d",&n);
printf("Enter reference string:\n");
for(i=0;i<n;i++)
scanf("%d",&pages[i]);
printf("Enter number of frames: ");
scanf("%d",&f);
for(i=0;i<f;i++)
frames[i]=-1;
printf("\nOptimal Page Replacement\n");
for(i=0;i<n;i++)
{
found=0;
for(j=0;j<f;j++)
{
if(frames[j]==pages[i])
{
found=1;
```

```
break;
}
}
if(found==0)
{
pageFaults++;
pos=-1;
for(j=0;j<f;j++)
{
if(frames[j]==-1)
{
pos=j;
break;
}
}
if(pos==-1)
{
farthest=-1;
for(j=0;j<f;j++)
{
nextUse=n+1;
for(k=i+1;k<n;k++)
{
if(frames[j]==pages[k])
{
nextUse=k;
```

```
break;
}
}
if(nextUse>farthest)
{
farthest=nextUse;
pos=j;
}
}
}
frames[pos]=pages[i];
}
printf("%d : ",pages[i]);
for(k=0;k<f;k++)
{
if(frames[k]==-1)
printf("- ");
else
printf("%d ",frames[k]);
}
if(found==0)
printf("Page Fault");
else
printf("Hit");
printf("\n");
}
printf("\nTotal Page Faults = %d\n",pageFaults);
return 0;
```

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OUTPUT:

TEST CASE 1:

```
071A6225\Optimal.exe
Enter number of pages: 5
Enter reference string:
 1 2 3 4 1
Enter number of frames: 3

Optimal Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
3 : 1 2 3 Page Fault
4 : 1 4 3 Page Fault
1 : 1 4 3 Hit

Total Page Faults = 4
```

TEST CASE 2:

```
071A6225\Optimal.exe
Enter number of pages: 5
Enter reference string:
 1 2 1 3 2
Enter number of frames: 3

Optimal Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
1 : 1 2 - Hit
3 : 1 2 3 Page Fault
2 : 1 2 3 Hit

Total Page Faults = 3
```

TEST CASE 3:

```
071A6225\Optimal.exe
Enter number of pages: 6
Enter reference string:
1 2 3 4 1 2
Enter number of frames: 3

Optimal Page Replacement
1 : 1 - - Page Fault
2 : 1 2 - Page Fault
3 : 1 2 3 Page Fault
4 : 1 2 4 Page Fault
1 : 1 2 4 Hit
2 : 1 2 4 Hit

Total Page Faults = 4
```